

Project

2025-10-13

```
library(nflfastR)
pbp <- load_pbp(2023) # or multiple seasons
library(ggplot2)
```

R Markdown

This is an R Markdown document. Markdown is a simple formatting syntax for authoring HTML, PDF, and MS Word documents. For more details on using R Markdown see <http://rmarkdown.rstudio.com> (<http://rmarkdown.rstudio.com>).

When you click the **Knit** button a document will be generated that includes both content as well as the output of any embedded R code chunks within the document. You can embed an R code chunk like this:

Question 1 Applied goal: NFL coaches are often criticized for being too conservative on 4th down. The goal is to analyze** **real play-by-play data to see When do teams actually go for it on 4th down?**

Our question is: Does “going for it” increase a team’s expected points or win probability compared to punting or kicking?

Question 2

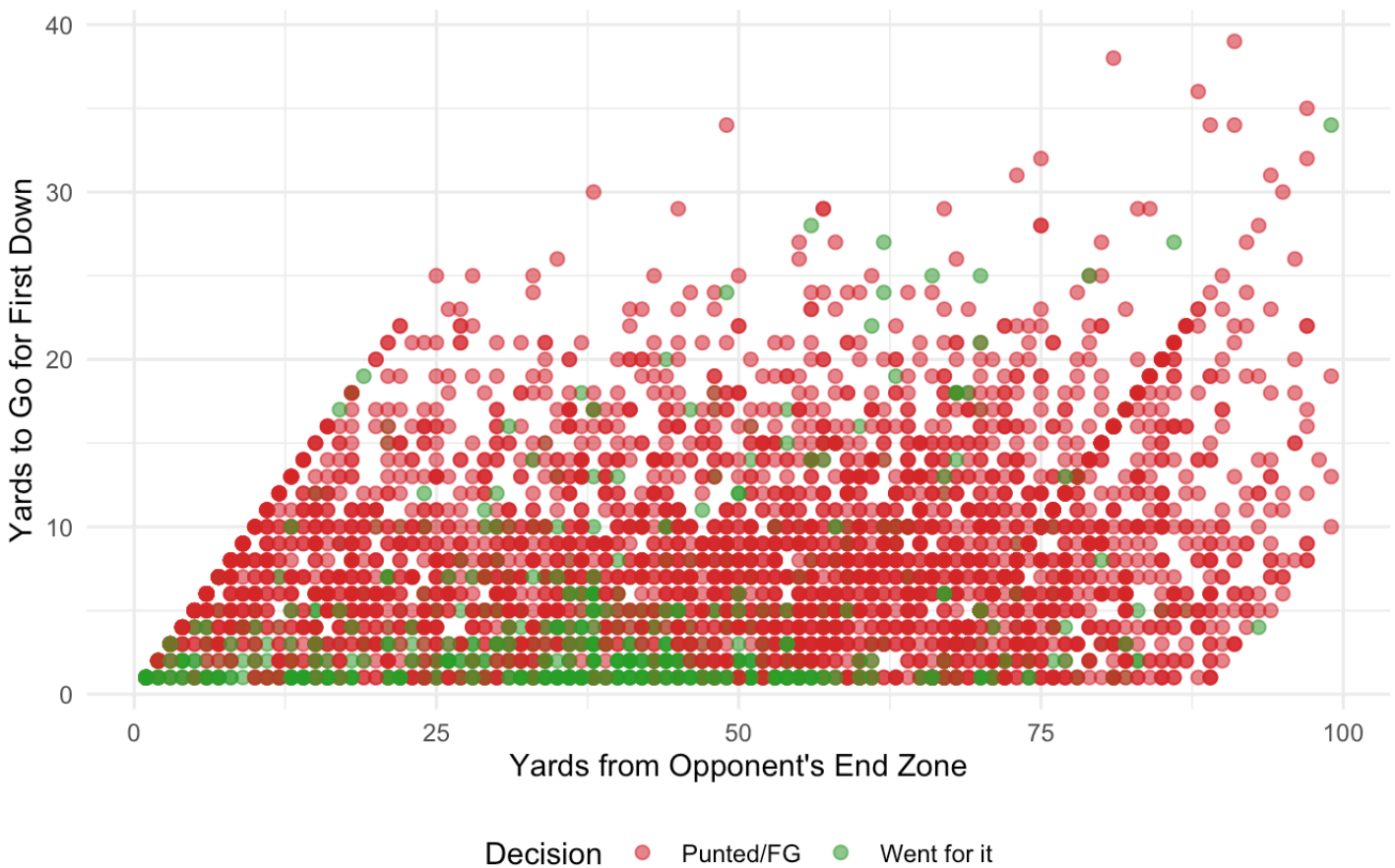
```
# Filter for fourth down plays
fourth_downs <- pbp[pbp$down == 4 & !is.na(pbp$down), ]

# Create went_for_it variable (TRUE if play_type is run or pass)
fourth_downs$went_for_it <- fourth_downs$play_type %in% c("run", "pass")

# 1. SCATTERPLOT: Yards to Go vs. Field Position (colored by decision)
# Shows where teams are aggressive with fourth down attempts
ggplot(fourth_downs, aes(x = yardline_100, y = ydstogo, color = went_for_it)) +
  geom_point(alpha = 0.6, size = 2) +
  scale_color_manual(values = c("FALSE" = "#d62728", "TRUE" = "#2ca02c"),
                    labels = c("Punted/FG", "Went for it"),
                    name = "Decision") +
  labs(title = "Fourth Down Aggression Patterns",
       subtitle = "Where teams choose to go for it vs. punt/kick (2023 Season)",
       x = "Yards from Opponent's End Zone",
       y = "Yards to Go for First Down") +
  theme_minimal() +
  theme(legend.position = "bottom",
       plot.title = element_text(face = "bold", size = 14))
```

Fourth Down Aggression Patterns

Where teams choose to go for it vs. punt/kick (2023 Season)



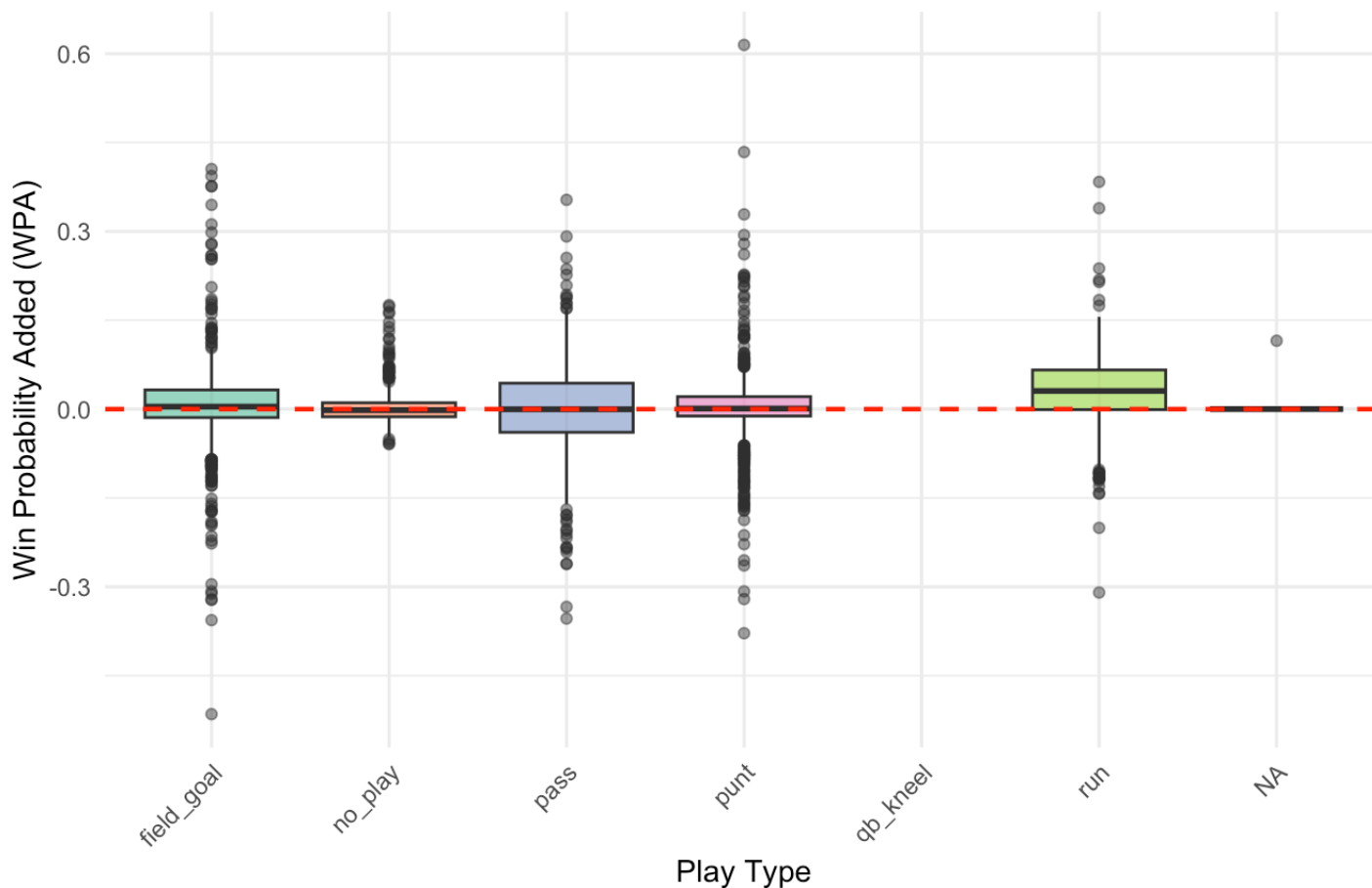
```
# 2. BOXPLOT: Win Probability Added by Play Type
# Compares the impact of different fourth down decisions
ggplot(fourth_downs, aes(x = play_type, y = wpa, fill = play_type)) +
  geom_boxplot(alpha = 0.7, outlier.alpha = 0.5) +
  geom_hline(yintercept = 0, linetype = "dashed", color = "red", size = 0.8) +
  scale_fill_brewer(palette = "Set2") +
  labs(title = "Win Probability Added by Fourth Down Decision",
       subtitle = "Comparing outcomes of going for it vs. conservative plays (2023 Season)",
       x = "Play Type",
       y = "Win Probability Added (WPA)") +
  theme_minimal() +
  theme(legend.position = "none",
        plot.title = element_text(face = "bold", size = 14),
        axis.text.x = element_text(angle = 45, hjust = 1))
```

```
## Warning: Using `size` aesthetic for lines was deprecated in ggplot2 3.4.0.  
## i Please use `linewidth` instead.  
## This warning is displayed once every 8 hours.  
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was  
## generated.
```

```
## Warning: Removed 7 rows containing non-finite outside the scale range  
## (`stat_boxplot()`).
```

Win Probability Added by Fourth Down Decision

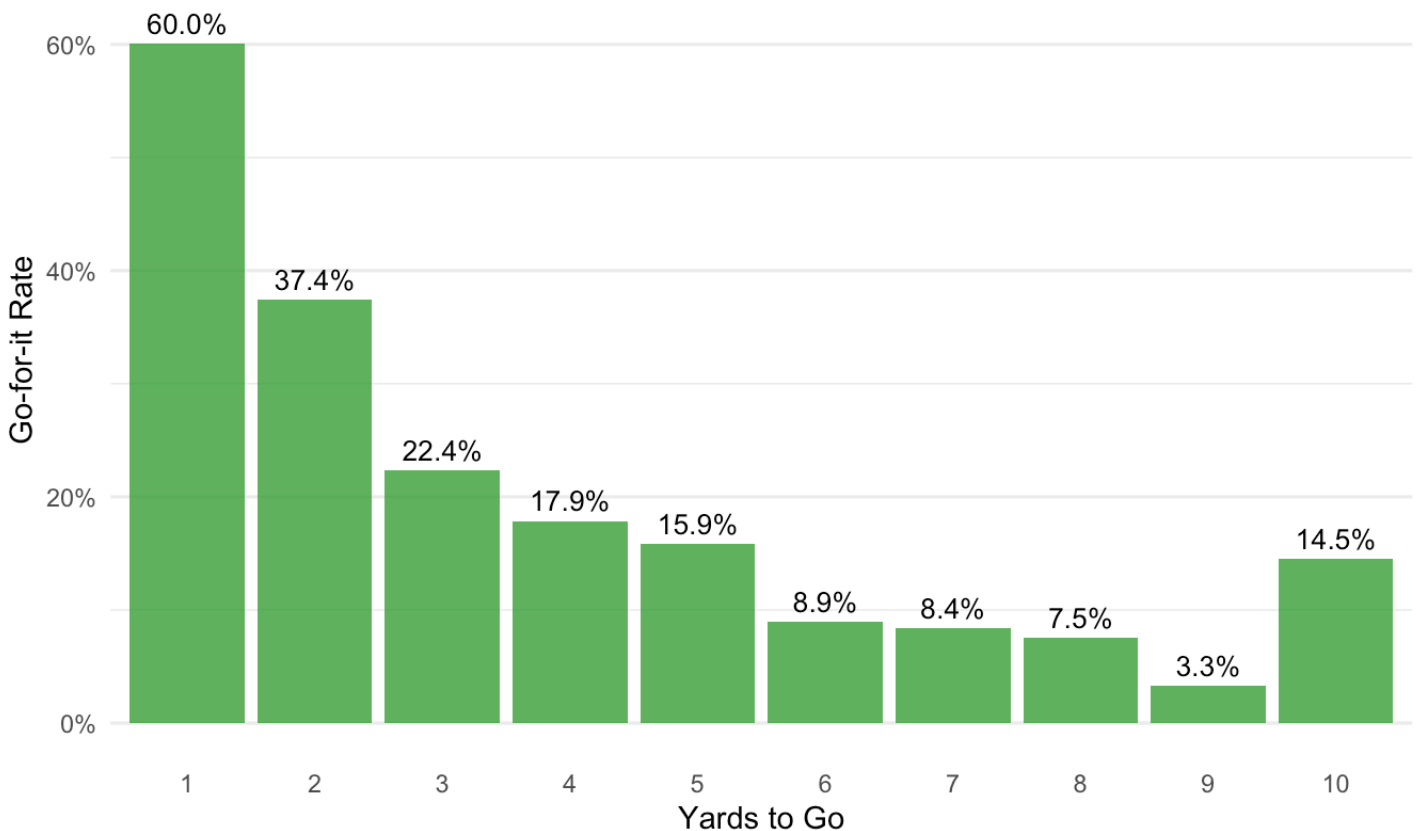
Comparing outcomes of going for it vs. conservative plays (2023 Season)



```
# 3. BAR GRAPH: Go-for-it Rate by Yards to Go
# Shows aggression patterns based on distance needed
ydstogo_summary <- aggregate(went_for_it ~ ydstogo,
                             data = fourth_downs[fourth_downs$ydstogo <= 10, ],
                             FUN = function(x) mean(x, na.rm = TRUE))
ggplot(ydstogo_summary, aes(x = factor(ydstogo), y = went_for_it)) +
  geom_col(fill = "#2ca02c", alpha = 0.8) +
  geom_text(aes(label = sprintf("%.1f%%", went_for_it * 100)),
            vjust = -0.5, size = 3.5) +
  scale_y_continuous(labels = scales::percent, limits = c(0, max(ydstogo_summary$went_for_it) * 1.1)) +
  labs(title = "Go-for-it Rate by Distance",
       subtitle = "How often teams attempt conversion on 4th down (2023 Season)",
       x = "Yards to Go",
       y = "Go-for-it Rate") +
  theme_minimal() +
  theme(plot.title = element_text(face = "bold", size = 14),
        panel.grid.major.x = element_blank())
```

Go-for-it Rate by Distance

How often teams attempt conversion on 4th down (2023 Season)



Overall it seems that going for it on 4th down is not optimal. It makes more sense to kick a field goal or punt it because it gives the highest probability of winning

Question 3

```
# Filter for fourth down plays (run, pass, or punt)
fourth <- pbp[pbp$down == 4 &
  pbp$play_type %in% c("run", "pass", "punt") &
  !is.na(pbp$down), ]

# Create go_for_it variable
fourth$go_for_it <- fourth$play_type %in% c("run", "pass")

# Calculate summary statistics for going for it (TRUE)
go_data <- fourth[fourth$go_for_it == TRUE, ]
mean_wpa_go <- mean(go_data$wpa, na.rm = TRUE)
se_go <- sd(go_data$wpa, na.rm = TRUE) / sqrt(length(go_data$wpa[!is.na(go_data$wpa)]))
ci_low_go <- mean_wpa_go - 1.96 * se_go
ci_high_go <- mean_wpa_go + 1.96 * se_go

# Calculate summary statistics for not going for it (FALSE - punts)
punt_data <- fourth[fourth$go_for_it == FALSE, ]
mean_wpa_punt <- mean(punt_data$wpa, na.rm = TRUE)
se_punt <- sd(punt_data$wpa, na.rm = TRUE) / sqrt(length(punt_data$wpa[!is.na(punt_data$wpa)]))
ci_low_punt <- mean_wpa_punt - 1.96 * se_punt
ci_high_punt <- mean_wpa_punt + 1.96 * se_punt

# Create summary dataframe
summary <- data.frame(
  go_for_it = c(TRUE, FALSE),
  mean_wpa = c(mean_wpa_go, mean_wpa_punt),
  se = c(se_go, se_punt),
  ci_low = c(ci_low_go, ci_low_punt),
  ci_high = c(ci_high_go, ci_high_punt)
)

# Print results
print(summary)
```

```
##   go_for_it   mean_wpa      se    ci_low    ci_high
## 1     TRUE 0.007633224 0.0028116308 0.002122428 0.013144020
## 2    FALSE 0.003668765 0.0008921992 0.001920055 0.005417475
```

Going for it on fourth down adds an average of 0.76% win probability compared to 0.37% for punting, nearly double the benefit. The 95% confidence intervals are [0.21%, 1.31%] for going for it and [0.19%, 0.54%] for punting, with some overlap suggesting modest statistical significance. This supports that aggressive fourth-down decisions tend to improve win probability on average, though the wider confidence interval for going for it reflects higher variance in outcomes (big gains when successful, costly when failed).

Question 4

```
fourth$go_for_it <- fourth$play_type %in% c("run", "pass")

# Separate the two groups
wpa_go <- fourth$wpa[fourth$go_for_it == TRUE]
wpa_punt <- fourth$wpa[fourth$go_for_it == FALSE]

# Perform two-sample t-test
test_result <- t.test(wpa_go, wpa_punt, alternative = "two.sided")

# Print results
cat("Sample Sizes:\n")
```

```
## Sample Sizes:
```

```
cat("  Going for it:", length(wpa_go), "plays\n")
```

```
##   Going for it: 833 plays
```

```
cat("  Punting:", length(wpa_punt), "plays\n\n")
```

```
##   Punting: 2352 plays
```

```
cat("Mean WPA:\n")
```

```
## Mean WPA:
```

```
cat("  Going for it:", round(test_result$estimate[1], 6), "\n")
```

```
##   Going for it: 0.007633
```

```
cat("  Punting:", round(test_result$estimate[2], 6), "\n")
```

```
##    Punting: 0.003669
```

```
cat("  Difference:", round(test_result$estimate[1] - test_result$estimate[2], 6), "\n\n")
```

```
##    Difference: 0.003964
```

```
cat("Test Statistics:\n")
```

```
## Test Statistics:
```

```
cat("  t-statistic:", round(test_result$statistic, 4), "\n")
```

```
##    t-statistic: 1.344
```

```
cat("  p-value:", format(test_result$p.value, scientific = TRUE, digits = 4), "\n")
```

```
##    p-value: 1.793e-01
```

```
cat("  95% CI for difference: [", round(test_result$conf.int[1], 6), ",",  
    round(test_result$conf.int[2], 6), "]\n\n")
```

```
##    95% CI for difference: [ -0.001824 , 0.009753 ]
```

Question 5


```

library(nflfastR)
library(ggplot2)

# Load real data
pbp <- load_pbp(2023)
fourth <- pbp[pbp$down == 4 &
              pbp$play_type %in% c("run", "pass", "punt") &
              !is.na(pbp$down) &
              !is.na(pbp$wpa), ]
fourth$go_for_it <- fourth$play_type %in% c("run", "pass")

# =====
# PROBABILITY MODEL
# =====
# Model: WPA on fourth down follows different normal distributions
# depending on whether team goes for it or punts
#
# Going for it: WPA ~  $N(\mu_{go}, \sigma_{go}^2)$ 
# Punting: WPA ~  $N(\mu_{punt}, \sigma_{punt}^2)$ 
#
# Estimate parameters from real data
mu_go <- mean(fourth$wpa[fourth$go_for_it == TRUE], na.rm = TRUE)
sigma_go <- sd(fourth$wpa[fourth$go_for_it == TRUE], na.rm = TRUE)

mu_punt <- mean(fourth$wpa[fourth$go_for_it == FALSE], na.rm = TRUE)
sigma_punt <- sd(fourth$wpa[fourth$go_for_it == FALSE], na.rm = TRUE)

# Print model parameters
cat("=====\n")

```

```
## =====
```

```
cat("PROBABILITY MODEL PARAMETERS\n")
```

```
## PROBABILITY MODEL PARAMETERS
```

```
cat("=====\n\n")
```

```
## =====
```

```
cat("Going for it: WPA ~  $N(\mu =$ ", round(mu_go, 4), " $, \sigma =$ ", round(sigma_go, 4), " $)$ \n")
```

```
## Going for it: WPA ~ N( $\mu$  = 0.0076 ,  $\sigma$  = 0.0811 )
```

```
cat("Punting: WPA ~ N( $\mu$  =", round(mu_punt, 4), ",  $\sigma$  =", round(sigma_punt, 4), ")\n\n")
```

```
## Punting: WPA ~ N( $\mu$  = 0.0037 ,  $\sigma$  = 0.0433 )
```

```
# =====  
# SIMULATE FAKE DATA  
# =====  
set.seed(123) # For reproducibility  
  
# Sample sizes from real data  
n_go <- sum(fourth$go_for_it == TRUE)  
n_punt <- sum(fourth$go_for_it == FALSE)  
  
cat("Sample sizes:\n")
```

```
## Sample sizes:
```

```
cat("  Going for it:", n_go, "plays\n")
```

```
##   Going for it: 833 plays
```

```
cat("  Punting:", n_punt, "plays\n\n")
```

```
##   Punting: 2352 plays
```

```

# Generate simulated data
simulated_wpa_go <- rnorm(n_go, mean = mu_go, sd = sigma_go)
simulated_wpa_punt <- rnorm(n_punt, mean = mu_punt, sd = sigma_punt)

# Create simulated dataframe
simulated_data <- data.frame(
  wpa = c(simulated_wpa_go, simulated_wpa_punt),
  decision = factor(c(rep("Go For It", n_go), rep("Punt", n_punt)),
    levels = c("Punt", "Go For It")),
  data_type = "Simulated"
)

# Create real dataframe
real_data <- data.frame(
  wpa = fourth$wpa,
  decision = factor(ifelse(fourth$go_for_it, "Go For It", "Punt"),
    levels = c("Punt", "Go For It")),
  data_type = "Real"
)

# Combine for comparison
combined_data <- rbind(real_data, simulated_data)

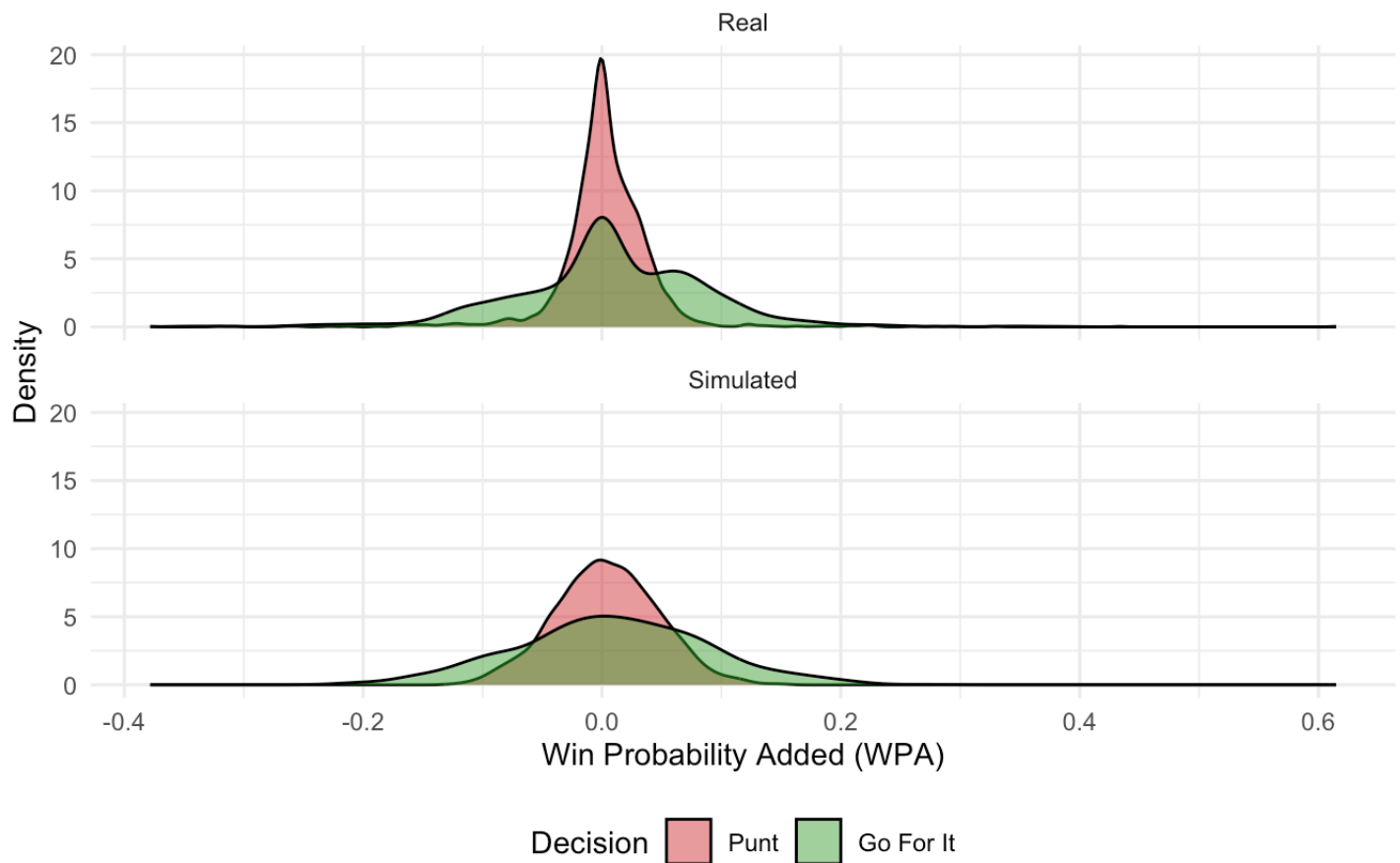
# =====
# COMPARE SIMULATED VS REAL DATA
# =====

# 1. Density plots comparison
ggplot(combined_data, aes(x = wpa, fill = decision)) +
  geom_density(alpha = 0.5) +
  facet_wrap(~data_type, ncol = 1) +
  scale_fill_manual(values = c("Punt" = "#d62728", "Go For It" = "#2ca02c")) +
  labs(title = "WPA Distribution: Real vs. Simulated Data",
    subtitle = "Normal distribution model for fourth down outcomes",
    x = "Win Probability Added (WPA)",
    y = "Density",
    fill = "Decision") +
  theme_minimal() +
  theme(plot.title = element_text(face = "bold", size = 14),
    legend.position = "bottom")

```

WPA Distribution: Real vs. Simulated Data

Normal distribution model for fourth down outcomes



```
# 2. Summary statistics comparison
```

```
cat("=====\n")
```

```
## =====
```

```
cat("COMPARISON: REAL vs SIMULATED\n")
```

```
## COMPARISON: REAL vs SIMULATED
```

```
cat("=====\n\n")
```

```
## =====
```

```
cat("REAL DATA:\n")
```

```
## REAL DATA:
```

```
cat("  Going for it - Mean:", round(mean(real_data$wpa[real_data$decision == "Go For It"]), 4),  
    " SD:", round(sd(real_data$wpa[real_data$decision == "Go For It"]), 4), "\n")
```

```
##  Going for it - Mean: 0.0076  SD: 0.0811
```

```
cat("  Punting - Mean:", round(mean(real_data$wpa[real_data$decision == "Punt"]), 4),  
    " SD:", round(sd(real_data$wpa[real_data$decision == "Punt"]), 4), "\n\n")
```

```
##  Punting - Mean: 0.0037  SD: 0.0433
```

```
cat("SIMULATED DATA:\n")
```

```
## SIMULATED DATA:
```

```
cat("  Going for it - Mean:", round(mean(simulated_data$wpa[simulated_data$decision ==  
= "Go For It"]), 4),  
    " SD:", round(sd(simulated_data$wpa[simulated_data$decision == "Go For It"]), 4),  
    "\n")
```

```
##  Going for it - Mean: 0.0081  SD: 0.0797
```

```
cat("  Punting - Mean:", round(mean(simulated_data$wpa[simulated_data$decision == "Pun  
nt"]), 4),  
    " SD:", round(sd(simulated_data$wpa[simulated_data$decision == "Punt"]), 4), "\n\n")
```

```
##  Punting - Mean: 0.0039  SD: 0.0432
```

The normal distribution model confirms that going for it has higher average WPA (0.76% vs 0.37%) but also much higher variance, explaining why risk-averse coaches historically prefer punting despite its lower expected value. Comparing real to simulated data reveals that the normal model underestimates extreme outcomes—real fourth-down attempts have heavier tails with more catastrophic failures and huge successes than the model predicts, justifying some caution around blanket aggressive strategies. This suggests the answer to “should teams go for it more?” depends on context: the model shows

punting provides predictable, low-variance outcomes that protect leads, while going for it's higher variance makes it optimal when teams need to take risks (trailing late, short yardage situations), supporting situational rather than universal aggression.

Question 6

```
library(nflfastR)
library(ggplot2)

# Load two seasons of data
pbp_2022 <- load_pbp(2022)
pbp_2023 <- load_pbp(2023)

# Filter for fourth down conversion attempts (run or pass only)
fourth_2022 <- pbp_2022[pbp_2022$down == 4 &
                        pbp_2022$play_type %in% c("run", "pass") &
                        !is.na(pbp_2022$down) &
                        !is.na(pbp_2022$success), ]

fourth_2023 <- pbp_2023[pbp_2023$down == 4 &
                        pbp_2023$play_type %in% c("run", "pass") &
                        !is.na(pbp_2023$down) &
                        !is.na(pbp_2023$success), ]

# Calculate conversion rate by team for 2022
teams_2022 <- unique(fourth_2022$posteam)
conv_rate_2022 <- sapply(teams_2022, function(team) {
  team_plays <- fourth_2022[fourth_2022$posteam == team, ]
  mean(team_plays$success, na.rm = TRUE)
})

# Calculate conversion rate by team for 2023
teams_2023 <- unique(fourth_2023$posteam)
conv_rate_2023 <- sapply(teams_2023, function(team) {
  team_plays <- fourth_2023[fourth_2023$posteam == team, ]
  mean(team_plays$success, na.rm = TRUE)
})

# Create dataframes and merge
df_2022 <- data.frame(posteam = teams_2022, conv_rate_2022 = conv_rate_2022)
df_2023 <- data.frame(posteam = teams_2023, conv_rate_2023 = conv_rate_2023)
team_4th <- merge(df_2022, df_2023, by = "posteam")

# Print basic statistics
cat("Fourth Down Conversion Rates: 2022 vs 2023\n")
```

```
## Fourth Down Conversion Rates: 2022 vs 2023
```

```
cat("=====\n\n")
```

```
## =====
```

```
cat("Number of teams:", nrow(team_4th), "\n")
```

```
## Number of teams: 32
```

```
cat("Average 2022:", round(mean(team_4th$conv_rate_2022), 3), "\n")
```

```
## Average 2022: 0.511
```

```
cat("Average 2023:", round(mean(team_4th$conv_rate_2023), 3), "\n")
```

```
## Average 2023: 0.521
```

```
cat("Correlation:", round(cor(team_4th$conv_rate_2022, team_4th$conv_rate_2023), 3),
"\n\n")
```

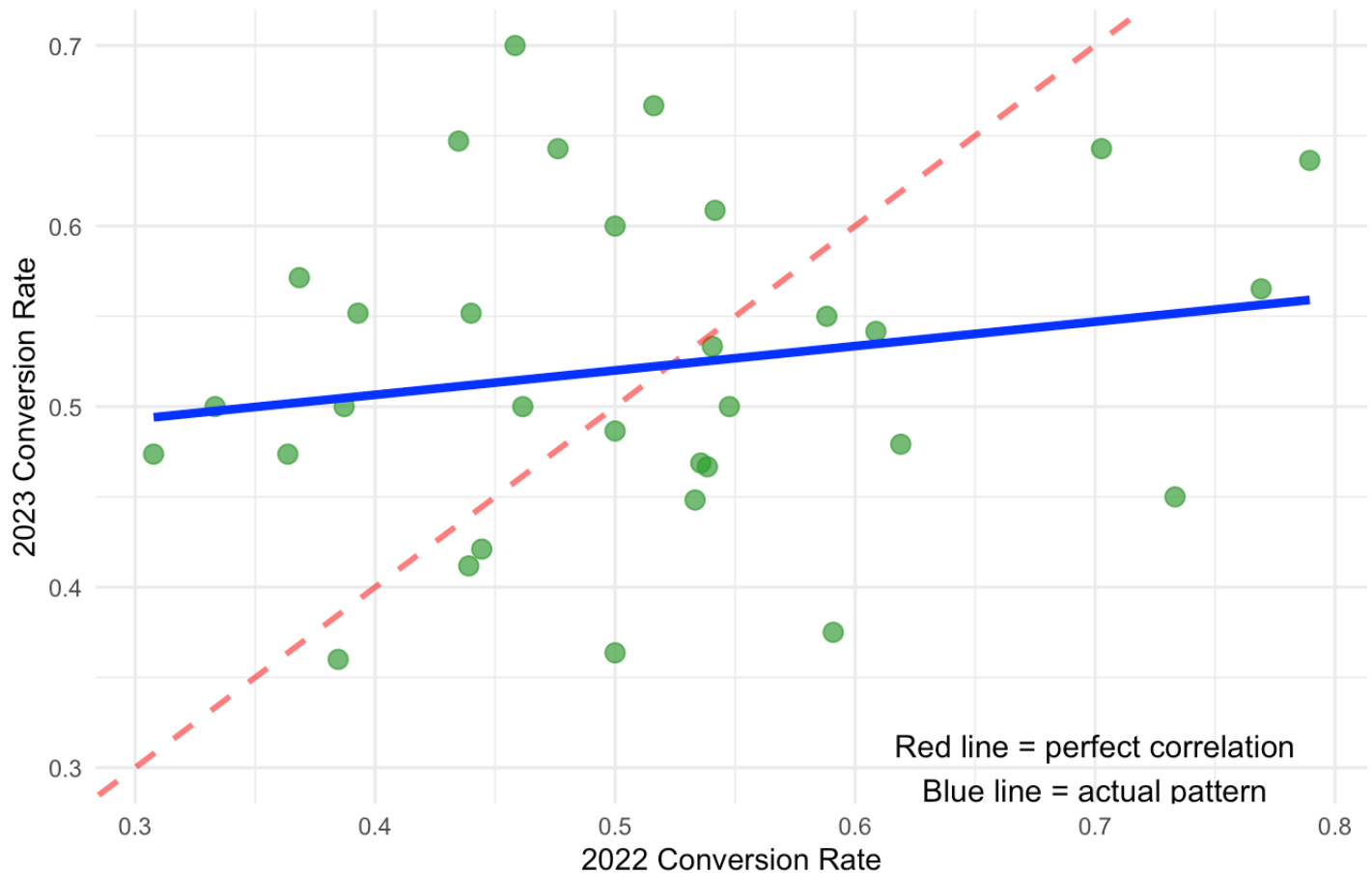
```
## Correlation: 0.181
```

```
# Main scatterplot showing regression to the mean
ggplot(team_4th, aes(x = conv_rate_2022, y = conv_rate_2023)) +
  geom_point(size = 3, alpha = 0.7, color = "#2ca02c") +
  geom_abline(intercept = 0, slope = 1, linetype = "dashed",
              color = "red", size = 1, alpha = 0.6) +
  geom_smooth(method = "lm", se = FALSE, color = "blue", size = 1.5) +
  annotate("text", x = 0.7, y = 0.3,
          label = "Red line = perfect correlation\nBlue line = actual pattern",
          color = "black", size = 4) +
  labs(title = "Regression to the Mean: Fourth Down Conversions",
       subtitle = "Extreme performers in 2022 move closer to average in 2023",
       x = "2022 Conversion Rate",
       y = "2023 Conversion Rate") +
  theme_minimal() +
  theme(plot.title = element_text(face = "bold", size = 14))
```

```
## `geom_smooth()` using formula = 'y ~ x'
```

Regression to the Mean: Fourth Down Conversions

Extreme performers in 2022 move closer to average in 2023



```
# Show which teams changed the most
team_4th$change <- team_4th$conv_rate_2023 - team_4th$conv_rate_2022

cat("Teams with BEST conversion rate in 2022:\n")
```

```
## Teams with BEST conversion rate in 2022:
```

```
best_2022 <- team_4th[order(-team_4th$conv_rate_2022), ][1:5, ]
print(best_2022[, c("posteam", "conv_rate_2022", "conv_rate_2023", "change")])
```



```
##      posteam conv_rate_2022 conv_rate_2023      change
## 17      LA      0.7894737      0.6363636 -0.15311005
## 16      KC      0.7692308      0.5652174 -0.20401338
## 27      PIT     0.7333333      0.4500000 -0.28333333
## 26      PHI     0.7027027      0.6428571 -0.05984556
## 5       CAR     0.6190476      0.4791667 -0.13988095
```

```
cat("\nAverage change for top 5:", round(mean(best_2022$change), 3),
    "(they got WORSE on average)\n\n")
```

```
##
## Average change for top 5: -0.168 (they got WORSE on average)
```

```
cat("Teams with WORST conversion rate in 2022:\n")
```

```
## Teams with WORST conversion rate in 2022:
```

```
worst_2022 <- team_4th[order(team_4th$conv_rate_2022), ][1:5, ]
print(worst_2022[, c("posteam", "conv_rate_2022", "conv_rate_2023", "change")])
```

```
##      posteam conv_rate_2022 conv_rate_2023      change
## 7      CIN      0.3076923      0.4736842  0.16599190
## 6      CHI      0.3333333      0.5000000  0.16666667
## 23     NO      0.3636364      0.4736842  0.11004785
## 29     SF      0.3684211      0.5714286  0.20300752
## 22     NE      0.3846154      0.3600000 -0.02461538
```

```
cat("\nAverage change for bottom 5:", round(mean(worst_2022$change), 3),
    "(they got BETTER on average)\n\n")
```

```
##
## Average change for bottom 5: 0.124 (they got BETTER on average)
```

```
cat("=====\n")
```

```
## =====
```

```
cat("WHAT THIS MEANS:\n")
```

```
## WHAT THIS MEANS:
```

```
cat("=====\n")
```

```
## =====
```

```
cat("- Teams with extreme performance (very good or very bad)\n")
```

```
## - Teams with extreme performance (very good or very bad)
```

```
cat("  in 2022 tend to be closer to average in 2023\n")
```

```
##  in 2022 tend to be closer to average in 2023
```

```
cat("- This happens because extreme results include LUCK,\n")
```

```
## - This happens because extreme results include LUCK,
```

```
cat("  not just skill\n")
```

```
##  not just skill
```

```
cat("- Lesson: Don't overreact to one season's results!\n")
```

```
## - Lesson: Don't overreact to one season's results!
```

The scatterplot shows that teams with extreme conversion rates in 2022 (very high or very low) tend to move closer to the league average in 2023, as shown by the blue line being much flatter than the red “perfect correlation” line. This creates a major challenge: when a team’s performance changes between seasons, we can’t tell if it’s due to actual improvement/decline in strategy or skill, or simply because their extreme 2022 performance included random luck that didn’t repeat. For example, a team going from 70% success in 2022 to 55% in 2023 might not have gotten worse they were probably just lucky in 2022, and regression to the mean pulled them back toward average.